Represent Rational Numbers and Their Opposites on the Number Line

Find the opposite of a fraction or decimal and place both on a number line.

Name _____ **Date** _____

Tonight's spark

A weather chart lists Monday's low as −5.6 degrees. A reporter asks for the opposite reading: the same distance from 0, on the other side. What is it?

Guess first: will the opposite reading be above 0 or below 0?

Our class model: the same distance, the other side

Try it

Work it out on paper. Write one answer on each line.

1 **What is the opposite of 3/4?**

A. 4/3 **B.** 3/4 **C.** −3/4 **D.** −4/3

My answer _____

2 **What is the opposite of −5.6?**

A. 5.6 **B.** −5.6 **C.** −6.5 **D.** 0.56

My answer _____

3 **Which pair are opposites?**

A. 1.2 and 2.1 **B.** 1.2 and 12 **C.** −1.2 and −2.1 **D.** 1.2 and −1.2

My answer _____

4 **The point −2.5 is 2.5 units left of 0. How far from 0 is its opposite?**

A. 5 units **B.** 0 units **C.** 2.5 units **D.** −2.5 units

My answer _____

5 **What is the opposite of the opposite of −6.35?**

A. 6.35 **B.** −6.35 **C.** 0 **D.** −63.5

My answer _____

6 **A point is halfway between 1 and 2. What is its opposite?**

A. −1.5 **B.** 1.5 **C.** −2.5 **D.** −0.5

My answer _____

Talk about it

Is the opposite of a fraction or decimal the same distance from 0? Which side is it on?

Home mission

Put a fork on the table as zero. Put a coin one hand-width to the right of the fork. Ask your student: where does the opposite coin go? Then ask: where does the opposite of the opposite go?

If we get stuck, ask

- Which two whole numbers does this value fall between?
- Same distance from zero — which side now?
- What happens if you take the opposite twice?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/7-2-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it